PW EARNERS • CODING & WEB DEVELOPMENT

# Lists & Tables

HTML ke Structural Elements — Lecture 05

Summary Notes | By – Nishant Sir

## 1. Lists in HTML

HTML mein **3 tarah ki lists** hoti hain, jo use hoti hain data ko organized tarike se dikhane ke liye:

Unordered List | Ordered List | Description List

| List Type | Tag | Kab use karein |
|---|---|---|
| Unordered List | `<ul>` | Jab order/sequence matter nahi karta — jaise shopping items, kisi bhi order mein likho farak nahi padta. |
| Ordered List | `<ol>` | Jab order/sequence matter karta hai — jaise recipe ke steps (pehle pyaaz kaato, phir tel garam karo). |
| Description List | `<dl>` | Jab kisi term ka naam aur uski explanation deni ho — jaise glossary, FAQs, definitions. |

### 1.1 Unordered List — <ul>

`<ul>` poori list ka **container** hai, aur `<li>` (List Item) har individual item ke liye hota hai jo iske andar hota hai.

> *<ul> ek dabbe jaisa hai jisme <li> items rakhe jaate hain — <ul> khud kuch nahi dikhata, uske andar ke <li> hi dikhte hain.*

```
<ul>
  <li>Hindi</li>
  <li>English</li>
  <li>Maths</li>
</ul>
```

### 1.2 Ordered List — <ol>

`<ol>` mein bhi `<li>` hi use hota hai — sirf container tag `<ul>` se `<ol>` mein badal jaata hai, aur items automatically numbered (1, 2, 3...) ho jaate hain.

```
<ol>
  <li>Pyaaz kaato</li>
  <li>Tel garam karo</li>
</ol>
```

### 1.3 Description List — <dl>

`<dl>` poori description list ka container hai. Isme do tags hote hain:

| Tag | Full Form | Kaam |
|---|---|---|
| `<dt>` | Description Term | Term ya title hota hai |
| `<dd>` | Description Details | Us term ki description/explanation hoti hai |

> *<dl> ek dabbe jaisa hai jisme <dt> terms aur unki <dd> descriptions rakhi jaati hain — <dl> khud kuch nahi dikhata, uske andar ke <dt> aur <dd> hi dikhte hain.*

```
<dl>
  <dt>HTML</dt>
  <dd>HyperText Markup Language</dd>
</dl>
```

### 1.4 List Styling — CSS list-style-type

Unordered list ke bullet ka style change karne ke liye `list-style-type` property use hoti hai:

| Value | Description |
|---|---|
| `disc` | Default filled circle |
| `circle` | Hollow circle |
| `square` | Filled square |
| `none` | Koi bullet point nahi |

### 1.5 Nested Lists

Lists ko **ek dusre ke andar** bhi likha ja sakta hai — jaise ek main list item ke andar poori sub-list. Yeh **nested list** kehlati hai.

**EXAMPLE — NESTED ORDERED + UNORDERED**

1. Fruits
    1. Mango
    2. Orange
2. Vegetables
    1. Cabbage
    2. Capsicum
        - Green Capsicum
        - Yellow Capsicum
        - Red Capsicum

👉 Yahaan outer list **ordered** hai, aur sabse andar wali (Capsicum ke colours) **unordered** hai.

### 📝 Homework — Nested List Practice

Ek deep nested list banao is structure ke saath:

I. List Item 1 → a, b (alphabetic)
II. List Item 2 → 1, 2 (numeric) → 2.2 ke andar bullets → usme bhi square bullets
III. List Item 3 → simple bullets

Hint: `type="a"`, `type="1"` attributes aur nested `<ul>`/`<ol>` ka combination use karo.

# 2. Tables in HTML

## 2.1 List vs Table — Kab kya use karein?

| Situation | Use karein |
|---|---|
| Data ek hi dimension mein ho (sirf items, ek ke baad ek) | **List** |
| Data grid mein ho (rows aur columns dono) | **Table** |

Table basically **rows aur columns ka collection** hota hai.

## 2.2 Table ke Core Tags

| Tag | Full Form | Kaam |
|---|---|---|
| `<table>` | Table | Poori table ka container (jaise poori marksheet ka frame) |
| `<tr>` | Table Row | Ek horizontal row |
| `<td>` | Table Data | Ek individual cell (box), row ke andar |

```
<table>
  <tr>
    <td>90</td>
    <td>50</td>
  </tr>
</table>
<!-- Emmet shortcut: table>tr*2>td*3 -->
```

## 2.3 Table ke Structural Tags — thead, tbody, th

| Tag | Full Form | Kaam |
|---|---|---|
| `<thead>` | Table Head | Header rows ko group karta hai |
| `<tbody>` | Table Body | Actual data rows ko group karta hai |
| `<th>` | Table Header | Column ya row ka title/label (bold + center by default) |

**EXAMPLE — MARKS TABLE**

| | Maths | Hindi | English | Science |
|---|---|---|---|---|
| **Sita** | 90 | 50 | 70 | 30 |

## 2.4 Caption

`<caption>` table ke upar ek title/heading dikhane ke liye use hota hai — jaise "A test table with merged cells".

## 2.5 Merging Cells — colspan & rowspan

| Attribute | Kaam |
|---|---|
| `colspan` | Ek cell ko horizontally (left-right) multiple **columns** jitna bada banata hai |
| `rowspan` | Ek cell ko vertically (top-bottom) multiple **rows** jitna bada banata hai |

**EXAMPLE — MERGED CELLS TABLE**

*A test table with merged cells*

| | Average | | Red eyes |
|---|---|---|---|
| | height | weight | |
| **Males** | 1.9 | 0.003 | 40% |
| **Females** | 1.7 | 0.002 | 43% |

👉 "Average" header `colspan="2"` se do columns (height + weight) cover karta hai. Blank corner cell aur "Red eyes" header `rowspan="2"` se do rows cover karte hain.

### 📝 Theek-Thaak Homework — Time Table

Ek weekly Time Table banao jisme:

- Top row mein **"Time Table"** heading — poori width mein colspan se merged
- Left side mein **"Hours"** label — rowspan se saari rows cover kare
- Mon–Fri columns, beech mein ek **"Lunch"** row jo poori width cover kare
- Friday ke last do periods mein **"Project"** cell — rowspan se merged

### 📝 Crazy Homework — Seminar Schedule

Ek advanced nested-merge table banao:

- **Day** column — Monday, Tuesday, Wednesday rows ko rowspan se group karo (jitne slots utni height)
- "Seminar" ek top-level header — poori width colspan
- Uske neeche "Schedule" (Begin + End columns) aur "Topic" column
- Monday: 2 topics (2 rows), Tuesday: 3 alag time-slots (3 rows), Wednesday: 1 row

Yeh homework **colspan + rowspan dono ko ek saath, multiple levels mein** use karne ki practice karwata hai.

### ✅ Quick Recap

- **Lists:** <ul> (unordered), <ol> (ordered), <dl>+<dt>+<dd> (description) — sabme <li> item hota hai (dl chhod ke)
- **Nested Lists:** ek list dusri list ke andar — different types combine kar sakte hain
- **Tables:** <table> → <tr> (row) → <td>/<th> (cell)
- **Structure:** <thead> (header rows), <tbody> (data rows)
- **Merging:** colspan = columns merge, rowspan = rows merge
- **Caption:** table ka title dikhane ke liye

Coding and Web Development — Lecture 05 | Notes by Claude, based on Nishant Sir's class